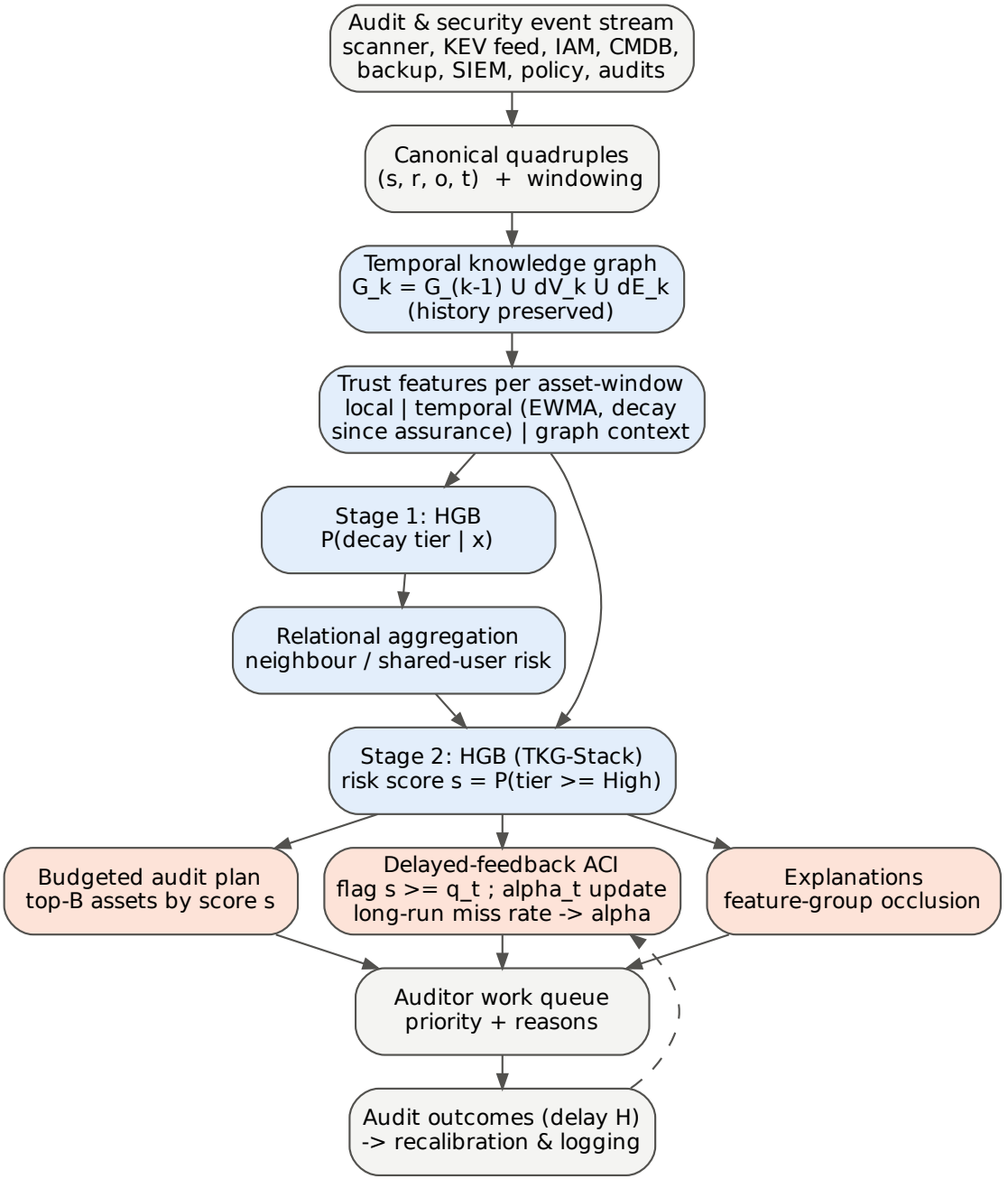
Audit & security event stream
scanner, KEV feed, IAM, CMDB,
backup, SIEM, policy, audits
Canonical quadruples
(s, r, o, t) + windowing
Temporal knowledge graph
G_k = G_(k-1) U dV_k U dE_k
(history preserved)
Trust features per asset-window
local | temporal (EWMA, decay
since assurance) | graph context
Stage 1: HGB
P(decay tier | x)
Relational aggregation
neighbour / shared-user risk
Stage 2: HGB (TKG-Stack)
risk score s = P(tier >= High)
Budgeted audit plan
top-B assets by score s
Delayed-feedback ACI
flag s >= q_t ; alpha_t update
long-run miss rate -> alpha
Explanations
feature-group occlusion
Auditor work queue
priority + reasons
Audit outcomes (delay H)
-> recalibration & logging